

1.0.0 Initial Stage Gate Plan for Thracian Webgui V3

Stage Name: Core Functionality / MVP Build

Stage Duration: 40-60 hours

Stage Gate Meeting Date: Tuesday, November 6th, 2024 at 12:00PM

Objective: Establish core functionality for mission creation, management, and transmission.

1. Stage Goals and Objectives

Build core features that allow users to create and manage relative/absolute missions.

Implement basic mission controls, including Play, Pause, and Stop.

Ensure reliability through robust testing.

Document major paradigms & important instructions for future developers and other team members to reference.

2. Scope of Work

Mission Management

- CRUD operations for missions

- Support for relative time mission type

- Support for absolute time mission type

Task Management

- Add or remove tasks within missions

- Support for tasks using relative time

- Support for tasks using absolute time

Transmission Controls

- Start Button

- Pause/Resume Button

- Stop Button

- Repeat Button

- Show current progress

- Resume progress on refresh

Error handling & Security

- Out of bounds power inputs

- Name length validation

- Input sanitation

- Invalid / Empty inputs error handling

- Websockets connection interruption

- File format type validation

Styling & UX

- Basic animations

- Consistent / professional look & feel

- Live system status updates

- Dynamically ordered tasks for absolute missions

- Buttons dynamically disabled

3. Key Deliverables

Release-ready code backed by testing results that demonstrate stability.

4. Quality Criteria and Testing

Unit Tests:

- Mission CRUD
 - Missing inputs
 - Invalid inputs
- Task Create + Destroy
 - Missing inputs
 - Invalid inputs
- Transmitter Start, Pause/play, Stop
 - With and without replay enabled
 - Mission progression on refresh

Integration Tests

- Start newly created missions
- Start missions with newly uploaded iq files
- Start missions with newly uploaded wav files
- Run absolute missions for multiple days
- Websocket connection interruption
- Odroid Integration
 - Apache server integration
 - Offline testing

5. Dependencies

Backend

- Must assign unique mission IDs
- Must assign unique task IDs per mission
- Must reliably update system status
- Must reliably update transmitter status
- Must have acceptable error handling

Development Tools: React, Redux

Testing Tools: Local Thracian with working backend, Postman for API testing

6. Risks and Mitigation

Risk: Additional to-be-discovered bugs

Mitigation: Begin development on lesser-understood features first, giving ample time for newly discovered issues to get resolved.

Risk: Rushed code leading to technical debt accrual

Mitigation: Communicate progress + challenges frequently allowing for adjustments of the acceptable amount of technical debt based on business needs.

7. Timeline and Milestones

Week 1:

- Transmission Controls
- Validate user inputs and handle major errors.

Week 2:

- Absolute Mission Support
- Unit Testing
- Styling

End of Stage:

- Integration testing
- Documentation
- Prepare for Stage Gate review meeting

8. Stage Gate Criteria for Moving Forward

- Functionality: All core features are fully operational.
- Code Quality: Code suitable for a wide array of additional features to be built upon.
- Testing: Critical tests passed with no major unresolved issues.
- Documentation: Setup instructions & developer insights documented.